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3 October 1986
SUPERSEDING
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28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWU (INCLUDING VARIATION LS2SWUA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be supplied in two variations: LS2SWU (unarmored) and LS2SWUA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First	- Copper conductor, AWG no. 18, class B stranding, (7 by 0.0152 inch) tin coated. Maximum diameter 0.050 inch.
Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	- Cross-linked polyethylene insulation. Nominal thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
Fourth	- Two conductors, one black and one white, cabled together with nominal lay of 3 inches to form each pair. For LS2SWU-1.1/
Fifth	- Binder tape over each pair, at manufacturer's option, applied helically with overlap.

1/ For LS2SWU-1, omit seventh, eighth and ninth.

(B) denotes changes.

AMSC N/A FSC 6145
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Sixth - Braided shield of AWG no. 34 or no. 36 tin coated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.

Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

- Diameter over completed pair (see table I).

Eighth - The required number of pairs (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward. Fillers shall be employed to form a firm round cable.

Ninth - Binder tape applied helically with overlap.

Construction variant for type LS2SWU

Tenth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LS2SWUA

(B) Tenth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking not required.

Eleventh - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/33	Size 2/	Number of pairs	Diameter over completed pair nominal (inch)	Cable jacket minimum (inch)	Overall diameter (inches)			
					Type LS2SWU		Type LS2SWUA	
					min	max	min	max
-01UN	LS2SWU-1	1	0.213	0.012	0.240	0.255	0.290	0.305
-02UN	LS2SWU-3	3	.213	.060	.670	.710	.720	.760
-03UN	LS2SWU-7	7	.213	.060	.860	.910	.910	.960
-04UN	LS2SWU-12	12	.213	.070	1.130	1.200	1.190	1.250
-05UN	LS2SWU-19	19	.213	.080	1.292	1.380	1.342	1.420
-06UN	LS2SWU-24	24	.213	.080	1.500	1.590	1.550	1.640
-07UN	LS2SWU-30	30	.213	.080	1.670	1.760	1.720	1.810
-08UN	LS2SWU-37	37	.213	.100	1.785	1.870	1.825	1.920
-09UN	LS2SWU-61	61	.213	.100	2.205	2.300	2.255	2.350

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.

2/ Type LS2SWU and LS2SWUA.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	7.47
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to shield 3/.....	500
Shield to water 4/.....	500
Group A:	
Visual and dimensional.....	No failure
Hydrostatic (open end) - leakage at 300 lb/in^2, 2 hours, in^3, maximum.....	1.0
Capacitance	
Mutual capacitance - at 1 kHz, pF/ft, maximum.....	30
Capacitance unbalance - percent, maximum.....	10
Characteristic impedance, nominal (at sinusoidal frequency of 1.0 MHz), ohms.....	75 ± 5
Cable attenuation	
Power loss (at sinusoidal frequency of 1.0 MHz), decibels (dB)/100 feet, maximum.....	1.5
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	80

3/ For LS2SWU-12 and larger sizes, test not less than 25 percent of the total quantity of shields in the core.

4/ Test all shields in LS2SWU-1 through -7. For LS2SWU-12 and larger sizes, test all shields that are in the outer periphery of the core.

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EXAMINATION AND TESTS: (Continued)

	Requirements
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - (type LS2SWU only) cycles, minimum	250
Cable filler removability (all types, except LS2SWU-1)	No failure
Armor - conformance to material, construction and coverage, (type LS2SWUA only)	No failure
Heat distortion - percent of unaged, maximum	30
Shield - conformance to material, construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Aging and compatibility (cable) (125 ± 5°C)	No failure
(B)	Abrasion resistance (jacket) - scrapes, minimum	75
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Flame propagation (cable)	No failure
	Immersion (jacket)	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged, minimum	50
	Smoke index, maximum	
	Jacket	25
	Fillers	--
(B)	Insulation	45
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:

Size	Feet (nominal)
1 through 12	1500
19 through 24	1000
30 through 61	500

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
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